Multi-Class Human Activity Recognition Using Engineered Smartphone Sensor Features

Farha Shaikh
BS Data Science and Applications
Indian Institute of Technology Madras

Abstract

This study investigates multi-class human activity recognition (HAR) using engineered accelerometer and gyroscope features from the UCI Human Activity Recognition dataset. Logistic Regression and Random Forest classifiers were trained to classify six physical activities. The Logistic Regression model achieved approximately 96% test accuracy and a mean weighted F1-score of 0.94 under stratified cross-validation. Feature importance analysis revealed gravity-based orientation features as dominant predictors for static postures, while gyroscope jerk signals contributed to dynamic activity recognition.

1 Introduction

Human Activity Recognition (HAR) aims to classify physical activities using sensor data collected from wearable or smartphone devices. Such systems are widely applied in healthcare monitoring, fitness tracking, and context-aware computing.

This work explores multi-class activity recognition using pre-engineered statistical features derived from smartphone accelerometer and gyroscope signals.

2 Dataset

The UCI Human Activity Recognition dataset contains sensor recordings from 30 subjects performing six activities:

- Walking
- Walking Upstairs
- Walking Downstairs
- Sitting
- Standing
- Laying

The dataset provides 561 engineered features extracted from time-domain and frequency-domain sensor signals.

3 Methodology

3.1 Preprocessing

The dataset was split into predefined training and test sets. Feature standardization was applied within a machine learning pipeline.

3.2 Models Evaluated

Two supervised learning models were evaluated:

- Logistic Regression (linear baseline)
- Random Forest (non-linear ensemble)

Model stability was assessed using 5-fold Stratified Cross-Validation on the training data.

4 Results

4.1 Test Set Performance

Model	Test Accuracy	CV Mean F1
Logistic Regression	0.96	0.94
Random Forest	0.93	–

Logistic Regression outperformed Random Forest, indicating that the engineered features are largely linearly separable.

4.2 Confusion Analysis

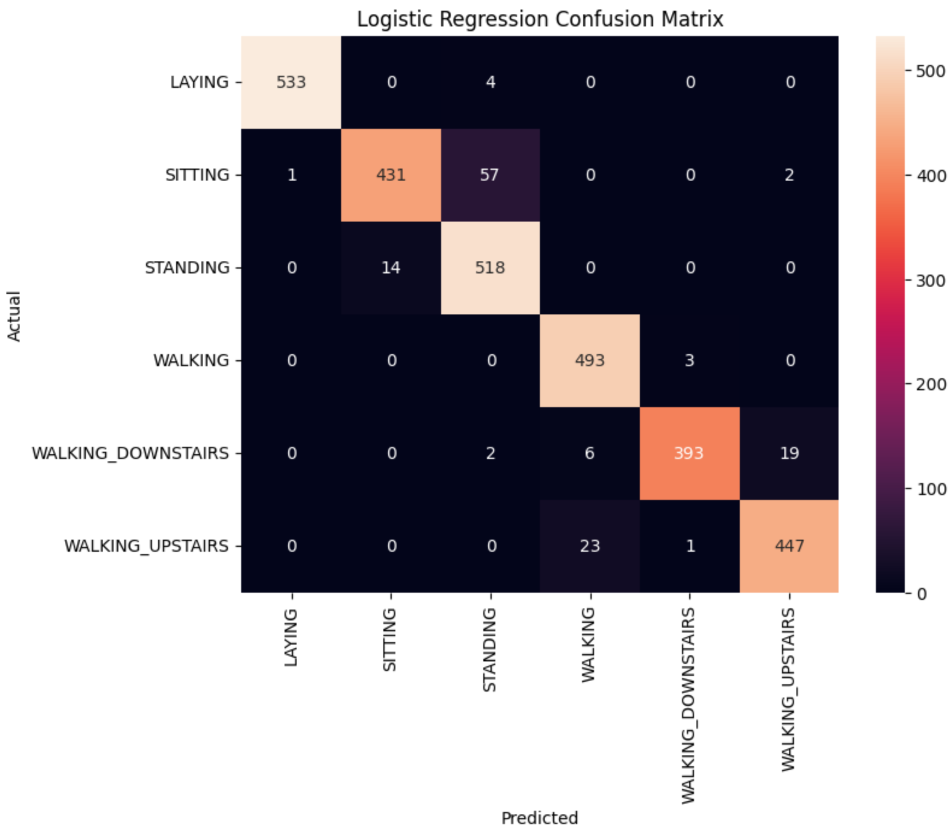

Figure 1: Confusion Matrix for Logistic Regression on the Test Set.

The confusion matrix revealed that Sitting and Standing exhibit the highest misclassification rates, likely due to similar gravity orientation signatures. Laying was classified with near-perfect accuracy due to distinct posture characteristics.

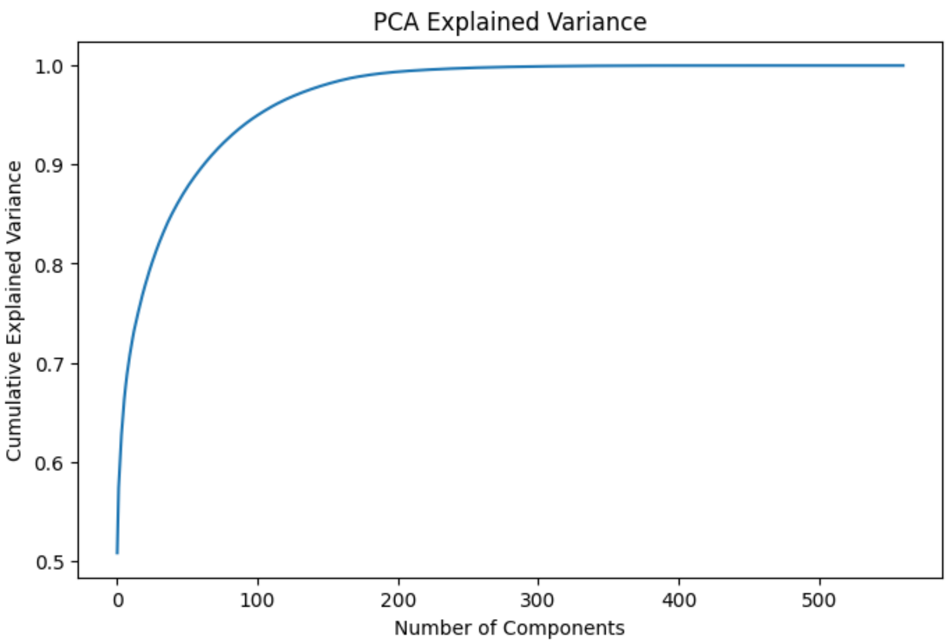

Figure 2: Cumulative Explained Variance as a Function of Principal Components.

4.3 Feature Importance

Analysis of Logistic Regression coefficients showed that:

- Gravity-based acceleration features dominate static posture classification.
- Gyroscope jerk and correlation features contribute significantly to dynamic motion detection.

5 Dimensionality Analysis

Principal Component Analysis (PCA) was applied to examine feature redundancy and dimensionality structure.

As shown in Figure 2, a reduced number of principal components explains the majority of variance, indicating substantial redundancy among the engineered features. Approximately 95% of variance is retained using a significantly smaller subset of components.

6 Conclusion

This study demonstrates that multi-class human activity recognition can be effectively performed using engineered statistical sensor features and simple linear models. The results highlight the importance of gravity orientation signals for posture recognition and motion-derived features for dynamic activities.

Future work may explore raw time-series modeling and lightweight edge deployment strategies.